

#Practice qs on Abstraction

1. An e-commerce app should accept different payments method (UPI, Credit card, Debit card, Netbanking) the app should not care the payment mode. create an abstraction to check through the payments.

2. Calculator Operations System.

Create an abstract class Calculator with abstract method calculate(). the derived classes should be

Addition, Subtraction, Multiplication and Division

Each derived class must implement the method calculate() to perform operation.

the user should not know the calculation process they can only call the method calculate()

3. Food Delivery App (like Zomato)

Design a system:

Abstract class: DeliveryPartner

Abstract methods:

deliver_order()

calculate_time(distance)

Create subclasses:

BikeDelivery

CarDelivery

Each should calculate time differently based on speed.

4. Ride Booking System (like Uber)

Abstract class: Ride

Abstract methods:

calculate_fare(distance)

get_vehicle_type()

Subclasses:

Mini

Sedan

SUV

Each ride type has different per km rates.

5.Streaming Platform (like Netflix)

Abstract class: Subscription

Abstract methods:

get_price()

get_quality()

Subclasses:

Basic

Standard

Premium

6.E-commerce Shipping (like Amazon)

Abstract class: Shipping

Abstract methods:

calculate_cost(weight)

delivery_time()

Subclasses:

StandardShipping

ExpressShipping

7.Employee Management System

Abstract class: Employee

Abstract methods:

calculate_salary()

get_role()

Subclasses:

Manager

Developer

Intern

Different salary logic for each.

8. Smart Home System

Abstract class: SmartDevice

Abstract methods:

turn_on()

turn_off()

Subclasses:

Light

Fan

AC

9. Notification System

Abstract class: Notification

Abstract method:

send(message)

Subclasses:

EmailNotification

SMSNotification

PushNotification

10. Vehicle Rental System

Abstract class: Vehicle

Abstract methods:

rent_cost(days)

get_type()

Subclasses:

Bike

Car

Truck